CST542 - Computational Thinking

CST 542 (Computational Thinking) Syllabus

Course Information

Course Details

- **Course Title:** Computational Thinking
- **Course Number:** CST 542
- **Meeting Location:** Physical
- **Units:** 2

Course Description

An investigation of computing pedagogy through the lens of object-oriented programming. Students will explore computer science curriculum by designing and completing assignments, selecting appropriate content, and evaluating teaching efficacy.

Materials

- Course textbook: "Computational Thinking Education" by Siu-Cheung Kong and Harold Abelson

Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system

Course Objectives

At the end of this class, you should be able to:

1. Design effective computer science curriculum using object-oriented programming principles
2. Create and evaluate educational assignments that promote computational thinking
3. Assess teaching efficacy and student learning outcomes in computing education
4. Apply pedagogical theories to computer science instruction and curriculum development

Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

Assignment Kind	Value
Projects	35%
Exams	35%
Participation	30%

Projects

Projects will take the form of curriculum design exercises, educational assignment development, and pedagogical analysis assignments. Students will create focused teaching materials and evaluate instructional approaches for computer science concepts.

Exams

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

Participation

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups. Therefore, attendance is required, as well as participation in all in-class activities. These activities may take the form of in-class discussions, brainstorming sessions, stand-ups, as well as making active progress on projects during class time.

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.